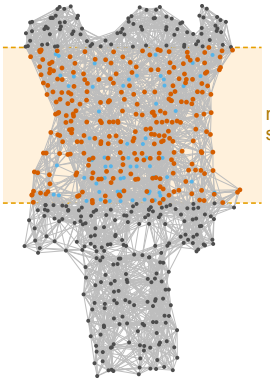
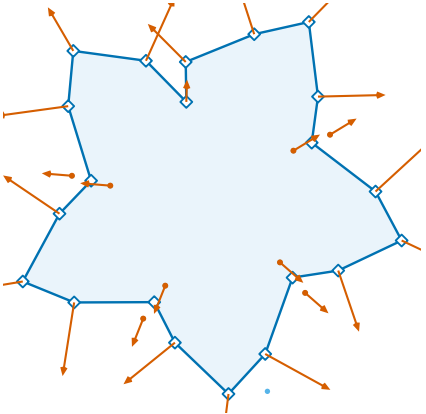
Cα outside the slab  membrane Cα  loaded outer shell  polygon vertex (shell mean)  nodal force $\tau\,\partial A/\partial \mathbf{x}$

**a** MscL Cα truss, $r_c$ = 10 Å




**b** one 4 Å layer, extracellular view



**c**

**Geometrically nonlinear truss**
$U = \frac{k}{2}\sum_e (\ell_e - \ell_e^0)^2$, exact kinematics

**Conservative membrane tension**
$\Pi = U - \tau A(\mathbf{x})$, $A = \frac{1}{2}\mathbf{x}^T\mathbf{H}\mathbf{x}$

**Gauge from the symmetries of the load**
$A$ invariant under 4 rigid motions; tilt carries a moment

**Verified, certified path**
Newton = VFIFE; inertia of $\mathbf{K}_t - \tau\mathbf{H}$ at every step

**Departure from linearization**
stiffening $a$, turning $\sin\theta \approx \chi$, snapping $n_{\mathrm{neg}}$

**Test against partner structures**
TM Cα RMSD vs. linear response and null models